# General Concepts

## General Terms

- **Bandwidth:** max data rate of a link (bits/sec).
- **Goodput:** proportion of successful slots (transmission without collision).
- **Throughput:** proportion of slots with a transmission (success or not).
- **Propagation time:** time for a signal to travel between two nodes.

## Mathematical Foundations

### Probability

- **Conditional:** $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$, $P(B) > 0$
- **Bayes:** $P(A \mid B) = \frac{P(B|A)P(A)}{P(B)}$
- **Total Prob. Law:** $P(B) = \sum_i P(B \mid A_i)P(A_i)$ (partition $\{A_i\}$)
- **Expected Value:** $\mathbb{E}[X] = \sum_x xP(X = x)$
- **Memoryless:** $P(X > s + t \mid X > t) = P(X > s)$ (Geo/Exp).

| Dist. | PMF | Notation | $\mathbb{E}[X]$ | $\mathrm{Var}(X)$ |
|---|---|---|---|---|
| Bernoulli | $P(X = 1) = p$ | $\mathrm{Ber}(p)$ | $p$ | $p(1-p)$ |
| Binomial | $\binom{n}{k}p^k(1-p)^{n-k}$ | $\mathrm{Bin}(n, p)$ | $np$ | $np(1-p)$ |
| Geometric | $(1-p)^{k-1}p$ | $\mathrm{Geo}(p)$ | $\frac{1}{p}$ | $\frac{1-p}{p^2}$ |
| Poisson | $\frac{\lambda^k e^{-\lambda}}{k!}$ | $\mathrm{Pois}(\lambda)$ | $\lambda$ | $\lambda$ |

- **Poisson Process:** events occur at rate $\lambda$; inter-arrival times $\sim \mathrm{Exp}(\lambda)$.
- In a Poisson process with rate $\lambda$, the probability of $k$ events in time $T$ is

$P_k(t = T) = \frac{(\lambda T)^k}{k!}e^{-\lambda T}$.

## Computer networks vs Distributed systems

### Circuit and Packet Switching

- **Circuit Switching:** establish a dedicated end-to-end path before sending.
  - Predictable performance/QoS once connected; data follows a fixed route.
  - Low per-packet overhead after setup; no per-packet path info needed.
  - If no free resources, the connection is blocked (no service).
  - Inefficient for bursty traffic: reserve peak bandwidth even if average is low.
  - All-or-nothing reliability: a failure breaks the circuit; must re-establish.
  - Setup delay: at least one RTT to establish the circuit.
- **Packet Switching:** no dedicated path; send data as packets over shared links.
  - Best-effort service; packets can be delayed, dropped, or arrive out of order.
  - Per-packet headers add overhead; requires reassembly at the destination.
  - Uses **statistical multiplexing** (assumption that traffic in network splits evenly over time): efficient for bursty traffic (no peak reservation).
  - Easier interconnection across organizations; more flexible routing.

  ***Note:*** *most modern networks (e.g., the Internet) use packet switching.*

### Multiplexing

- **Goal:** share one physical link among multiple circuits/flows.
- **TDM (Time Division):** divide time into frames, frames into slots; each flow gets fixed slots.
  - Requires tight synchronization and buffering so a sender has data when its slot arrives.
  - Predictable delay, but unused slots waste bandwidth (inefficient for bursty traffic).
- **FDM (Frequency Division):** split bandwidth into frequency bands; each flow gets a band.
  - No time synchronization, but each flow only uses a fraction of the spectrum.

## The OSI Model (5-Layer model)

- **Physical:** Move bits on a link (signals, timing, encoding).
- **Link:** Frames on local network, MAC addressing, error detection.
- **Network:** Packet delivery across networks, routing (IP).
- **Transport:** End-to-end process communication, reliability/flow control (TCP/UDP).
- **Application:** End-user services/protocols (DNS, DHCP, HTTP, FTP etc.).

# Link Layer & Local Area Networks (LAN)

## Shared Medium Intuition

- Nodes broadcast on same channel; only one can transmit at a time.
- Simultaneous transmit $\Rightarrow$ collision $\Rightarrow$ data lost.
- Sender uses full link bandwidth while transmitting.

## Definitions

- **Bandwidth:** max data rate of channel ($B$).
- **Throughput:** fraction of bandwidth used to send traffic.
- **Goodput:** fraction of bandwidth used for successful data.
- $\eta = \frac{\text{avg effective bandwidth}}{\text{total bandwidth}} = \frac{\text{avg effective time}}{\text{total time}}$, $0 \le \eta \le 1$.

## Multiple Access Domains

- **Multiple Access Domain:** set of nodes sharing a common channel.
- **Channel Partitioning:** divide channel into smaller "pieces" (time/freq/code).
- **Random Access:** transmit at will; handle collisions (ALOHA, CSMA).
- **Taking Turns:** nodes take turns to transmit (polling/token passing).

## ALOHA protocol

- Random access on shared medium; collisions if overlap. Frame time $T$ (size $L$, rate $B$): $T = L/B$.
- **Binomial model:** $N$ users, each transmits in a frame time $T$ w.p. $p$.
- **Poisson model:** Aggregate attempts are Poisson with rate $G$ (attempts per frame time $T$).
- **Slotted ALOHA** (align to slots of size $T$):
  - Success if exactly one transmission in a slot.
  - Binomial: $\mathbb{P}_{\text{suc}} = Np(1-p)^{N-1}$, max at $p^\star = 1/N$ gives max goodput (when $N$ is large) of $\eta \approx 1/e$.
  - Poisson: $\eta = Ge^{-G}$, Max $\eta_{\max} = 1/e$ at $G = 1$.
- **Pure ALOHA** (unslotted):
  - Vulnerable period $2T$ (current frame $\pm$ overlap).
  - Binomial (heuristic): $\mathbb{P}_{\text{suc}} \approx Np(1-p)^{2(N-1)}$, max at $p^\star = 1/(2N-1)$ gives max goodput (when $N$ is large) of $\eta \approx 1/(2e)$.
  - Poisson: $\eta = Ge^{-2G}$. Max $\eta_{\max} = 1/(2e)$ at $G = 0.5$.

## CSMA (Carrier Sense Multiple Access)

- **Basic Principle:** Listen (carrier sense) before transmitting.
- **Persistence Strategies:**
  - **1-persistent:** If idle, transmit immediately. If busy, wait until idle and transmit immediately (high collision probability).
  - **Non-persistent:** If idle, transmit. If busy, wait a random time (backoff) and check again.
  - **p-persistent:** If idle, transmit with probability $p$, else wait for next slot (compromise).
- **Propagation delay** $\tau$ ($T_{\text{prop}}$): Time for signal to travel between the two farthest nodes.

## CSMA/CD (Collision Detection) - Wired (Ethernet)

- **Mechanism:** Listen while transmitting. If collision detected (voltage spike), abort transmission, send **jam signal**, then backoff.
- **Condition for Detection:** Transmission time must be long enough to detect a collision from the farthest node and receive the collision signal back.

$$T_{\text{trans}} \ge 2 \cdot T_{\text{prop}}$$

(This determines the minimum packet size).
- **Example (minimum frame size):** distance $= 8\,\text{km}$, propagation speed $= 4 \times 10^7\,\text{m/s}$, rate $= 5\,\text{Mbps}$. Then

$$T_{\text{prop}} = \frac{8 \times 10^3}{4 \times 10^7} = 2 \times 10^{-4}\,\text{s} = 200\,\mu s \Rightarrow 2T_{\text{prop}} = 400\,\mu s$$

$$T_{\text{trans}} = \frac{S}{5 \times 10^6} \Rightarrow S = (5 \times 10^6)(4 \times 10^{-4}) = 2000\,\text{bytes}.$$

- **Binary Exponential Backoff:**
  - After $m$-th collision, choose $k$ uniformly from $\{0, \ldots, 2^m - 1\}$.
  - Wait $k \cdot$ slot time (where slot time $\approx 512$ bit times in Ethernet).
- **Efficiency** ($\eta$): As $N \to \infty$:

$$\eta = \frac{1}{1 + 2 \cdot \frac{T_{\text{prop}}}{T_{\text{trans}}} \cdot (e-1)}$$

## CSMA/CA (Collision Avoidance) - Wireless (WiFi/802.11)

- **Reasoning:** Cannot detect collisions reliably (weak received signal vs strong transmitted signal), so aim to avoid them.
- **Protocol:**
  - If channel idle for **DIFS**, transmit entire frame.
  - If busy, choose random backoff. **Timer counts down only when channel is idle** (freezes when busy).
  - Receiver sends **ACK** after **SIFS** if frame received successfully (no ACK = collision).
- **Virtual Carrier Sensing (RTS/CTS):** Prevents "Hidden Terminal" problem.
  - Sender transmits **RTS** (Request to Send).
  - Receiver replies **CTS** (Clear to Send).
  - Neighbors hear RTS/CTS and defer transmission (reserve the channel).

## Single (Logical) Link in practice

### MAC Addresses

- Link-layer address unique per NIC (Network Interface Card); assigned by IEEE OUI.
- Used to deliver frames on same LAN; flat (non-hierarchical), portable across LANs.
- Usually burned-in, but can be software-set.

### Ethernet

- Dominant wired LAN; many speed generations; cheap NICs.
- **CSMA/CD (classic shared medium):**
  1. Sense channel; if idle transmit, else wait.
  2. If collision detected while transmission, abort and send jam.
  3. **Binary exponential backoff:** after $m$-th collision pick $k \in \{0, \ldots, 2^m - 1\}$, wait $k \cdot 512$ bit-times, retry.

## Interconnecting broadcast domains

- **Switch:**
  - link-layer device that breaks broadcast domains; stores/forwards frames using MAC table.
  - Transparent to hosts; self-learning (no manual config), entries age out (TTL).
  - Each link is its own collision domain; enables simultaneous transmissions.
  - Forwarding: if dest known on incoming port $\Rightarrow$ drop; else forward to port; if unknown $\Rightarrow$ flood.
  - **Switch table:** (MAC, port, TTL). Learn from source MAC+ingress port on each frame.
- **Loops:** cause broadcast storms, multiple frame copies, MAC table instability.
- **Spanning Tree Protocol (STP):**
  - **Goal:** Prevent Layer 2 loops by logically blocking redundant paths; converges to a loop-free spanning tree.
  - **Mechanism:** Switches exchange **BPDUs** (Bridge Protocol Data Units) containing (RootID, Cost, SenderID, PortID).
  - **Algorithm Steps:**
    * **1. Elect Root Bridge:** Switch with the lowest **Bridge ID** becomes Root. (Initially, all claim to be Root).
    * **2. Select Root Ports (RP):** Each *non-root* switch selects **one** port with the best path to the Root.
    * **3. Select Designated Ports (DP):** Each *link/segment* selects **one** port to forward traffic (from the switch with the best path to Root).
    * **4. Block Others:** All non-RP and non-DP ports enter **Blocking** state (receive BPDUs, but drop data).
  - **Tie-Breakers (Lowest wins):**
    1. Lowest **Root Path Cost**.
    2. Lowest **Sender Bridge ID** (upstream switch).
    3. Lowest **Sender Port ID**.

## Error Detection & Correction

- **Key Relation:** To detect $d$ errors, need Hamming Distance $d_{min} \ge d + 1$. To correct $c$ errors, need $d_{min} \ge 2c + 1$.
- **Repetition (3x):** Send bit 3 times. Majority vote corrects 1 error. High overhead.
- **1D Parity:** Add 1 bit (odd/even). Detects odd # of errors. $d_{min} = 2$.
- **2D Parity:** Matrix of bits. Row+Col parities.
  - Detects up to 3 errors (guaranteed).
  - Fails to detect 4 errors if they form a rectangle.
  - Corrects 1 error (intersection of bad row/col).
- **Internet Checksum:** Sum 16-bit words (1's complement), append inverse. Used in IP/TCP. Detects most errors.

| Code | Overhead Ratio | $d_{min}$ | Can correct | Can detect |
|---|---|---|---|---|
| Repetition (3x) | $2/1$ (200%) | 3 | 1 | 2 |
| 1D Parity | $1/N$ | 2 | 0 | 1 |
| 2D Parity ($N \times N$) | $\approx 2/N$ | 4 | 1 | 3 |
| Internet Checksum | $16/L_{frame}$ | $\sim 2$ | 0 | Varies |

## "Layer 2.5" - ARP

ARP is generally considered to be part of Layer 2 (Data Link Layer), but it works directly to support Layer 3 (Network Layer).

- Why it's Layer 2: ARP messages are encapsulated directly into Ethernet frames. They do not have an IP header (Layer 3) and they cannot be routed outside of your local network (LAN).
- Why it feels like Layer 3: Its entire purpose is to handle Layer 3 information (IP addresses). It is the mechanism that allows Layer 3 to actually function on physical hardware.

### Address Resolution Protocol (ARP)

- Translates between MAC addresses and IP addresses on a LAN.
- Needed because Layer 3 provides the IP, but frames on Layer 2 need a destination MAC.
- To send outside the LAN, the packet uses the IP of the destination but the MAC of the next-hop router; routers then forward using their own L2 on each hop.
- Hosts use DHCP and routing to know whether the destination IP is on the same LAN or remote.
- Each node maintains an **ARP cache** (IP $\rightarrow$ MAC table).
- If the mapping is missing, send an **ARP request** (broadcast in the local broadcast domain).
- The target replies with an **ARP response** containing its MAC, which is stored in the cache.
- ARP messages stay within a single broadcast domain (not routed).

## Network Layer (IP & Routing)

### Introduction to IP addressing

- IPv4 address = 32 bits (a.b.c.d - 4 octets, each one 8 bits, ranging 0-255), identifies interface (host/router can have multiple interfaces).
- Subnet vs host split via CIDR notation: a.b.c.d/$x$ where $x$ = prefix length (subnet mask).
- Example: 192.168.1.178/24 $\Rightarrow$ subnet 192.168.1, host 178.
- The "Power of 2" Rule: adding 1 to the prefix length **cuts** the number of available host addresses in half.
- **Why IP is Hierarchical:** Unlike MAC addresses (flat/random), IP addresses are assigned topologically. This allows **Route Aggregation** (summarizing thousands of hosts into one table entry), which is critical for Internet scalability; ISPs advertise prefixes.
- **Router:** network-layer device that connects networks and forwards IP packets based on destination prefixes (routing table), decoupling L2 domains.
- **Router + ARP/L2:** for each hop, the router uses ARP on the outgoing interface to learn the next-hop MAC, rewrites the L2 header, and forwards the same IP packet (new L2 header per link).
- Forwarding uses **longest prefix match**.
- **Address discovery protocols:** DHCP (get IP config from server), ARP (map IP to MAC on LAN), DNS (map hostname to IP).

### IP Packet Format

- **Header** (3 32-bit rows): **[0-31]** Version(4), IHL(4), TOS(8), Total length(16) **[32-63]** Identification(16), Flags(3), Fragment offset(13) **[64-95]** TTL(8), Protocol(8), Header checksum(16)
- **Addresses:** Source address (32), Destination address (32)
- **Extra:** Options (0-40B) + Data (up to 65,515B)

### NAT (Network Address Translation)

- Translates internal private IPs to a smaller set of public IPs (often one) using port mapping.
- Internal hosts use private ranges (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16); NAT exposes a single public IP to the Internet.
- **Advantages:**
  - Conserves public IPv4 addresses; many hosts share one public IP.
  - Hides internal addressing and allows renumbering without changing external addresses.
  - Basic inbound shielding (unsolicited inbound blocked unless mapped).
- **How it works:**
  - NAT maintains a translation table mapping (private IP, private port) $\leftrightarrow$ (public IP, public port).
  - For outbound packets, NAT rewrites source IP:port and records mapping.
  - For inbound packets, NAT looks up the destination port and rewrites to the internal host; if no mapping, it drops.

  *Note: To allow connection to a specific port, we need to set it, as well as publish it. A server might use a **static NAT** mapping or **port forwarding** to allow presistent inbound access.*
- **Example setup:** LAN hosts A=192.168.0.1 (TCP ports 1024,2000), B=192.168.0.2 (UDP ports 1000,2000), C=192.168.0.3 (TCP port 1024). Public IP: 132.58.38.2. NAT allocates public ports 5001+.
- **Example table (public IP 132.58.38.2):**

| L4 Protocol | Translated L4 port | Translated IP | Original L4 port | Local IP |
|---|---|---|---|---|
| TCP | 5001 | 132.58.38.2 | 1024 | 192.168.0.1 |
| TCP | 5002 | 132.58.38.2 | 2000 | 192.168.0.1 |
| UDP | 5003 | 132.58.38.2 | 1000 | 192.168.0.2 |
| UDP | 5004 | 132.58.38.2 | 2000 | 192.168.0.2 |
| TCP | 5005 | 132.58.38.2 | 1024 | 192.168.0.3 |

- **Why it doesn't fully solve IPv4 shortage:**
  - Breaks end-to-end connectivity; inbound connections need port forwarding/ALGs.
  - Adds per-packet processing and state; can increase latency and complexity.
  - Scalability: still limited by public IPs for NAT gateways and port space.

### Topology

- Network graph structure (how nodes/links are connected).
- **Diameter:** max shortest-path distance between any two nodes (in hops/links).
- **Nodal degree:** number of links incident to a node.
- **Bisection bandwidth:** min total bandwidth across any split of nodes into two equal halves.
- **Edge expansion (intuition):** connectivity of small sets; higher expansion $\Rightarrow$ more robust.

| Topology ($N$ nodes) | Diameter | Degree | Bisection |
|---|---|---|---|
| Line | $N-1$ | 2 (ends 1) | 1 |
| Ring | $N/2$ | 2 | 2 |
| Balanced Tree | $2\log_2 N$ | 3 (root 2, leaves 1) | 1 |
| Hypercube ($N=2^k$) | $\log_2 N$ | $\log_2 N$ | $N/2$ |
| Complete Bipartite | 2 | $N/2$ | $N^2/8$ |

### Intradomain Routing

- Goal: choose a path for a packet through the network to reach the destination.
- **Shortest path routing:** pick the minimum-cost path (weights represent costs).
- **DV (Distance Vector):** each node knows only its neighbors and their distances (not full topology). Each node keeps a vector of distances to all destinations (anycast).

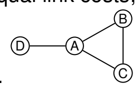
### Distance Vector (Bellman-Ford)

For node $s$:

$$d_s(v) = \begin{cases} c(s,v), & v \in \Gamma(s) \\ 0, & v = s \\ \infty, & \text{else} \end{cases}$$

Iterate:

- Update with neighbors: $d_s(v) = \min_{x \in \Gamma(s)}\{c(s,x) + d_x(v)\}$.
- If any entry changes, propagate updated vector.
- **Count-to-infinity:** when a path fails, nodes can keep increasing distances by looping updates.
- **Poisoned reverse:** if route from $X$ to $Z$ goes via $Y$, then $Y$ advertises $Z$ to $X$ with $\infty$ to prevent two-node loops (still fails for loops $\geq 3$).
  - Example (3-node loop): equal link costs; if link $A$–$D$ fails (weight $\infty$),




poisoned reverse still fails.
- **RIP (Routing Information Protocol):**
  - Distance-vector protocol using hop count as metric (max 15 hops).
  - Sends full distance vector every 30 seconds.
  - If no update for 180 seconds, route is considered unreachable (infinite).
- **ARPANET:** early distance-vector-like algorithm; suffered from slow convergence and required careful update control to avoid instability; not used today (historical).

### Link State (Dijkstra)

- Global knowledge: each node learns full topology and link costs; changes are flooded (multicast/LSA).
- Dijkstra from source $s$:

Initialize:

$$d_s(v) = \begin{cases} c(s,v), & v \in \Gamma(s) \\ 0, & v = s \\ \infty, & \text{else} \end{cases} \qquad N' = \{s\}$$

Repeat until $N' = N$:

1. Choose $w \notin N'$ with minimal $d_s(w)$.
2. Add $w$ to $N'$.
3. For each $v \in \Gamma(w) \setminus N'$, update $d_s(v) = \min\{d_s(v), d_s(w) + c(w,v)\}$.

- Once the table is built, forward using it (multicast routing supported).
- **OSPF (Open Shortest Path First):**
  - Link-state protocol; supports multiple metrics/weights, multiple paths, hierarchical areas.
  - Link costs/weights are configured by the network admin.
  - LSAs are flooded (multicast) on change; periodic refresh as well.
  - Each router builds full topology and runs Dijkstra locally.

### Comparison of LS vs DV

| Aspect | LS (Link State) | DV (Distance Vector) |
|---|---|---|
| Message complexity | $O(nE)$ (flood LSAs) | Neighbor exchanges only |
| Speed of convergence | Dijkstra $O(n^2)$; may oscillate | Varies; loops, count-to-infinity |
| Robustness | Wrong *link* cost affects own table | Wrong *path* cost propagates; blackholes |

### Traffic Engineering

- Goal: optimize network performance by tuning routing (often via link weights).
- Workflow: measure traffic/topology, build network-wide model, then optimize weights.
- Common objectives: maximize throughput, minimize congestion (max-link load), fairness.
  - **Minimize Congestion** - minimize the maximum link utilization: $\min \max_{e \in E} \frac{\text{load}(e)}{\text{capacity}(e)}$.
  - **Maximize Throughput** - maximize the total accepted traffic: $\max \sum_{(s,t)} \text{flow}(s,t)$.
  - **Maximum Multicommodity Flow (Max-MCF):** maximize total flow between all source-destination pairs subject to link capacities.
  - **Commodities** are tuples $(s,t,d)$ that represent demand $d$ from source $s$ to target $t$, e.g., $(A,B,12)$.
  - **Max-MCF$_{\text{OSPF/ECMP}}$:** maximize total flow using OSPF+ECMP (flows can split only on shortest paths).
  - **Max-MCF$_{\text{OPT}}$:** maximize total flow with arbitrary splitting (no shortest-path restriction).
  - **MinCong-MCF$_{\text{OSPF/ECMP}}$:** minimize max congestion using OSPF+ECMP.
  - **MinCong-MCF$_{\text{OPT}}$:** minimize max congestion with arbitrary splitting.

  *Note: In MinCong-MCF, all commodities must be sent (even if capacities are exceeded); in Max-MCF, flows are bounded by capacities.*
- **Complexity of link-weight optimization:**
  - NP-hard, even for simple objectives.
  - **Theorem:** Approximating the min-congestion multi-commodity flow within any constant factor is NP-hard, even for a single source-target pair.

### ECMP

- Equal-Cost Multi-Path: split traffic across multiple shortest paths (same total cost).
- Each router keeps a set of equal-cost next hops and load-balances among them.
- **Hashing**:
  - choose next hop by hashing the flow 5-tuple (src/dst IP, src/dst port, protocol).
  - This way we achieve deterministic per-flow mapping: all packets of a flow take the same path (avoids reordering).
  - Efficient to compute; provides near-even split across equal-cost paths in aggregate.

– Caveat: if only a few "heavy" flows exist, they may map to same path ⇒ imbalance.
- Good for load sharing, but optimality is not guaranteed (NP-hard).

# Transport Layer (TCP & UDP)

## TCP (Transmission Control Protocol)

- Reliable, in-order byte stream; receiver ACKs received data.
- **vs. UDP:** UDP provides port+IP multiplexing (best effort) with no reliability, ordering, or flow control.
- **Sequence #:** The byte number of the *first* byte of data in the segment.
- **ACK #:** The *next expected* byte (Cumulative ACK).
- **MTU vs. MSS:**
  – **MTU:** Max Transfer Unit (max packet size on link, e.g., 1500B).
  – **MSS:** Max Segment Size (max payload).
  $MSS = MTU - (IP_{Header} + TCP_{Header})$.
  Typically $MSS = MTU - 40$ (so $MSS < MTU$).
- **TCP header (min 20B, + options):** src/dst ports, Seq #, ACK #, flags (SYN/ACK/FIN/RST/PSH/URG), window, checksum, urgent pointer.

## Connection Setup (3-Way Handshake)

1. Client → Server: **SYN**, Seq $= x$.
   Announces client ISN (Initial Sequence Number, $x$) and requests connection.
2. Server → Client: **SYN + ACK**, Seq $= y$, ACK $= x + 1$.
   Accepts, advertises server ISN, and ACKs client ISN.
3. Client → Server: **ACK**, Seq $= x + 1$, ACK $= y + 1$.
   Final ACK; both sides synchronized.
4. (Connection Established)

## Connection Teardown

1. Client sends **FIN**.
2. Server replies **ACK**. (Client enters FIN_WAIT_2).
3. Server sends **FIN**.
4. Client replies **ACK**, waits **TIME_WAIT** (usually 2*MSL), then closes.

## Reliable data transfer (sender)

- Maintain sequence number and buffer; segment and send data.
- Send while window allows: $lastByteSent - lastByteAcked \le RW$ (receiver window).
- Start timer if not running; on timeout, retransmit the oldest unACKed segment.

## Reliable data transfer (receiver)

- Deliver in-order data and send cumulative ACK.
- If out-of-order segment arrives, send ACK for last in-order byte (duplicate ACK).

## Retransmissions

- Retransmit on either **timeout** or **duplicate ACKs**.
- **Fast retransmit:** many duplicate ACKs imply loss before timeout.

## RTT estimation

- Goal: set timeout slightly above RTT to avoid premature retransmits but still react fast to loss.
- **SampleRTT:** time from sending a segment until its first ACK returns (ignore retransmitted samples).
- Now we can compute a smoothed RTT which is exponentially weighted moving average of recent samples:
  $$\textbf{EstimatedRTT} \leftarrow (1-\alpha)\,\text{EstimatedRTT} + \alpha\,\text{SampleRTT}$$
  typical value for $\alpha$ is $1/8$ (higher $\alpha$ reacts faster; lower is smoother).
- RTT variation (we add some safety margin):
  $$\textbf{DevRTT} \leftarrow (1-\beta)\,\text{DevRTT} + \beta\,|\text{SampleRTT} - \text{EstimatedRTT}|$$
  typical value for $\beta$ is $1/4$ (higher $\beta$ reacts faster; lower is smoother).
- We may then set the timeout interval as:
  $$\textbf{TimeoutInterval} = \text{EstimatedRTT} + 4 \cdot \text{DevRTT}$$

## Flow Control vs. Congestion Control

- **Flow control**: prevents sender from overwhelming the receiver (receiver buffer limits).
- **Congestion control**: prevents sender from overloading the network.
- Sender can send while:
  $lastByteSent - lastByteAcked \le \min(RW, cwnd)$.
- When receiver buffer is the limiter, flow control dominates; when network is the limiter, congestion control dominates.

## Congestion Control Phases (AIMD)

**AIMD (Additive Increase, Multiplicative Decrease):** Ensures stability and fairness.

1. **Slow Start (SS):** Exponential growth; $cwnd = cwnd + 1$ per ACK (doubles every RTT).
2. **Congestion Avoidance (CA):** Linear growth; $cwnd = cwnd + 1/cwnd$ per ACK ($\approx +1$ MSS per RTT).

- **Transition:** Threshold **ssthresh** decides SS vs CA:
  If $cwnd < ssthresh$ → SS. If $cwnd \ge ssthresh$ → CA.

## TCP Tahoe

- On **any** loss (Timeout OR 3-Dup ACKs):
- Set $ssthresh = \lfloor cwnd/2 \rfloor$.
- Set $cwnd = 1$.
- Enter **Slow Start**.

## TCP Reno (Standard)

- Improves Tahoe for 3 duplicate ACKs (fast retransmit/fast recovery).
- On **Timeout:** Same as Tahoe ($cwnd = 1$, restart SS etc).
- On **3 Duplicate ACKs (Fast Retransmit/Recovery):**
  1. Retransmit missing segment immediately.
  2. Set $ssthresh = \lfloor cwnd/2 \rfloor$.
  3. Set $cwnd = ssthresh + 3$ (accounts for 3 packets buffering).
  4. Enter **Congestion Avoidance** (skips Slow Start).
- **Utilization intuition**
  – Reno provides sawtooth behavior in $cwnd$ over time.
  – With window size $W$ and RTT $RTT$, throughput is about $W/RTT$.
  – If window halves after loss, average throughput over a cycle is roughly $0.75\,W/RTT$.

## TCP Vegas

- Delay-based congestion control: compares expected throughput (from $cwnd/baseRTT$) to actual throughput (from $cwnd/RTT$).
- Adjusts $cwnd$ to keep the difference within a target range ($\alpha$, $\beta$), aiming to prevent queue buildup before loss.
- Less widely used because it competes poorly with loss-based flows (Reno), is sensitive to RTT measurement noise, and offers limited incremental deployability in the public Internet.

## UDP (User Datagram Protocol)

- Connectionless; no setup, no reliability, no ordering.
- Uses IP+port for multiplexing/demultiplexing (socket identified by destination port).
- Very small header; no congestion control, so can send as fast as the application wants.
- **UDP header (8B):** src/dst ports, length, checksum.
- Useful when timeliness matters more than reliability (e.g., streaming, VoIP), or when simplicity/overhead is key.
- Also used for request/response protocols like DNS and for DHCP (no TCP connection possible before address assignment).
- Reliability (if needed) is handled at the application layer.

## Reliable Data Transfer Protocols

These protocols are the conceptual building blocks for reliability. While TCP is a specific protocol, it uses the mechanisms defined in these models.

## Stop-and-Wait

- **Mechanism:** Sender sends one packet, waits for ACK; if ACK/packet lost, timeout triggers retransmission.
- **Efficiency:** Very low in "long fat pipes" (high RTT).
- **Transmission Delay:** $T_{packet} = L/R$ (where $L$=packet length, $R$=rate).
- **Utilization:** $U_{sender} = \frac{T_{packet}}{RTT + T_{packet}}$.
- **Timeout lower bound:** $T_{out} \ge 2T_{prop} + T_{ack} + T_{pt}$.
- $T_{prop} = \frac{distance}{propagation\ speed}$, $T_{ack} = \frac{ACK\ size}{rate}$,
  $T_{pt}$ = processing time.
- **Goodput (loss prob. $p$ for both data and ACK):**
  $$\text{Goodput} = \frac{T_{packet}}{\frac{1}{(1-p)^2}(T_{packet} + T_{out})} = \frac{T_{packet}(1-p)^2}{T_{packet} + T_{out}}.$$
  *Note: Here we assume independent losses for data and ACK packets. If ACK losses are negligible, replace $(1-p)^2$ with $(1-p)$ in the Goodput formula, since now it's Geometric with success prob. $(1-p)$.*
  *also, if $\overline{T_{out}}$ of successful transmissions is different than that of retransmissions, we would need to weight accordingly.*
  $$\text{Goodput} = \frac{T_{packet}}{\left(\frac{1}{(1-p)^2} - 1\right)(T_{packet} + T_{out}) + (T_{packet} + \overline{T_{out}})}$$

## Pipelined Protocols (Sliding Window):

Allows multiple unacknowledged packets to be "in-flight" to increase utilization.

## Go-Back-N (GBN):

- **Sender:** Can send up to $N$ packets without ACKs. Uses a *single timer* for the oldest unACKed packet.
- **Receiver:** Uses **Cumulative ACKs**. If a packet is missing, it discards all subsequent out-of-order packets.
- **Receiver window:** size 1; accepts only next in-order packet.
- **On Timeout:** Sender retransmits *all* $N$ transmitted but unACKed packets.
  *Note: if ACK for packet $k$ is lost, but ACK for $k+n$ arrives, sender can treat as if all packets up to $k+n$ were ACKed, sliding the window forward accordingly.*
- **Goodput (loss prob. $p$):** let $a = \frac{T_{out} + T_{packet}}{T_{packet}}$, then
  $\text{Goodput}_{GBN} = \frac{1-p}{1-p+pa}$.

## Selective Repeat (SR):

- **Sender:** Maintains a *timer for each* individual packet.
- **Receiver:** ACKs packets individually. **Buffers** out-of-order packets for later delivery.
- **On Timeout:** Sender retransmits *only* the specific lost packet.
- **Window Size Constraint:** To avoid sequence ambiguity,
  $WindowSize \le \frac{1}{2} \cdot MaxSeqNum$.

| Criteria | Selective Repeat | Go-Back-N |
|---|---|---|
| Bandwidth utilization | Retransmits only lost packets | May retransmit already received packets |
| Window sizes | Sender: $N$, Receiver: $N$ | Sender: $N$, Receiver: $1$ |
| Sequence numbers | $2N$ | $N+1$ |
| Out-of-order handling | Buffers and tracks out-of-order packets | No buffering; accepts only in-order |
| Receiver storage | Stores out-of-order until missing arrives | No need to store (sender may resend) |

## Implementation & Relation to TCP

- **Where they run:** These protocols run in the **Transport Layer** (Layer 4) within the **Operating System** of the end-hosts. Routers and switches (the "network core") generally do not implement these.
- **TCP Implementation:**
  – TCP is a hybrid: It uses **Cumulative ACKs** (like GBN) but typically **Buffers out-of-order segments** and can use **SACK** (Selective ACK) to retransmit only missing data (like SR).

# Application Layer (HTTP, DNS, SMTP & FTP)

## DNS (Domain Name System)

Maps between hostnames and IP addresses. Each server is authoritative for its domain and can resolve queries or forward them; resolution ends when an authoritative server replies or the requested data is found in cache.

- **RR Format** (Resource Record): $(Name, Value, Type, TTL)$.
- **Record types:**
  – **A**: maps an Internet hostname to an IP address.
  – **NS**: maps a domain name (e.g., `google.com`) to its responsible server.
  – **CNAME**: maps a hostname to its canonical name, e.g., `wedsac.mit.com` is actually `www.mit.com`.
  – **PTR**: reverse of A (IP address to hostname).
  – **MX**: maps a hostname to the mail server that handles it.
- **Server Hierarchy:**
  – **Local DNS (Resolver):** The first hop (ISP/Campus). Acts as a *Recursive Server* (performs the full lookup for the client) and caches results, according to received TTL (Time-To-Live).
  – **Root Server:** Top of the tree; directs queries to TLD servers (com, org).
  – **Authoritative Server:** The final server that knows the actual answer.

DNS queries and replies use UDP and typically port 53 (smaller packets for requests/replies).

**Advantages of DNS (names instead of IPs):**

1. Humans are not good at remembering IP addresses.
2. You can change a server's IP behind the scenes without users noticing.
3. You can map multiple IP addresses to one name for load balancing.

4. You can map multiple names to a single IP (e.g., `google.com` and `gmail.com`).

**Glued response:** if the client needs to query the authoritative server but does not know its IP, the server returns the IP in the NS response.

**Security issues:**

1. **DDoS Attack**: flooding DNS servers so they cannot answer other requests.
2. **MITM (Man-in-the-Middle) Attack**: when a client queries a server, a fake response can arrive before the real one, potentially mapping a requested name to a desired IP.
3. **Cache poisoning**: if the server is not well-protected, it continues serving an injected response until TTL expires; thus, we must ensure the reply is authentic.
4. **Kaminsky attack**: attacker sends many requests for random subdomains of a target domain (e.g., `random1.target.com`, `random2.target.com`, etc.) to a resolver; when the resolver queries the authoritative server, the attacker floods it with fake replies, hoping one matches the query ID and port, poisoning the cache.

- **DNSSEC**: adds signatures so you can verify who answered and add more security; adds extra information and expands UDP packet size.

### DHCP (Dynamic Host Configuration Protocol)

- Allows host to obtain IP dynamically (no manual config).
- Each time host connects, it can get a different IP; server can **lease** addresses and reuse later.
- Used by devices joining a network.
- **Message flow (DORA):**
  - **DHCP Discover:** src IP 0.0.0.0, dst 255.255.255.255 (broadcast, server unknown).
  - **DHCP Offer:** server replies, dst 255.255.255.255 (broadcast); includes **yiaddr** (your IP).
  - **DHCP Request:** client broadcasts 0.0.0.0 $\to$ 255.255.255.255 to accept one offer; includes chosen **yiaddr** so other servers withdraw.
  - **DHCP ACK:** server confirms lease; may broadcast so others know address in use.
- Each DHCP message includes a unique **Transaction ID (XID)** generated by the client; even if replies are broadcast, only the matching client accepts them.
- **DHCP relaying:** if DHCP server not on local subnet, relay forwards messages to server on another network.
  - Relay agent is usually the local router; client $\leftrightarrow$ relay uses broadcast, relay $\leftrightarrow$ server uses unicast.
  - Relay inserts **giaddr** (gateway IP) so server knows the client subnet and selects an address accordingly.
  - Relay forwards Discover/Request and returns Offer/ACK back to the client (continue with DORA).
- **Example (relayed Discover/Offer steps):**
  - Setup: H1 on LAN-A, router R0 (relay agent) connects LAN-A to LAN-B; DHCP server is on LAN-B. Example IPs: R0(LAN-A)=223.1.1.3, R0(LAN-B)=223.1.2.2.

| Message | SRC MAC | DST MAC | SRC IP | DST IP |
|---|---|---|---|---|
| DHCP Discover | MAC(H1) | Broadcast | 0.0.0.0 | 255.255.255.255 |
| Discover (giaddr=R0) | MAC(R0) | MAC(DHCP) | IP(R0) | IP(DHCP) |
| DHCP Offer (223.1.1.x) | MAC(DHCP) | MAC(R0) | IP(DHCP) | IP(R0) |
| DHCP Offer (223.1.1.x) | MAC(R0) | Broadcast | IP(R0) | 255.255.255.255 |

  - After the offer, the protocol continues similarly with Request and ACK.
- **Why it doesn't fully solve IPv4 shortage:**
  - **Peak demand**: works only if simultaneously active devices $<$ pool size; if everyone connects at once, the pool still runs out.
  - **Temporary fix**: improves reuse but does not increase the global IPv4 address space ($2^{32}$).

## Interdomain Routing & BGP

### Autonomous Systems (AS)

- **Definition:** IP networks/routers under one admin with a common routing policy.
- **ASN:** Unique 16/32-bit AS identifier used in BGP.
- **AS relationships:**
  - **Customer / Provider:** Customer pays provider for transit.
  - **Full / Partial transit:** Full = global reachability; partial = limited reach.
  - **Peering:** Settlement-free; advertise own + customers only.
- **AS-level topology:** Common categories
  - **Tier-1** ($\sim 10$ ASes, e.g. AT&T): No providers; global reach via peering.
  - **Tier-2:** Providers + peers (buys some transit).
  - **Stub AS** ($85\%$ of ASes): Single provider; no downstream customers.

### Border Gateway Protocol (BGP)

- **Definition:** Inter-AS routing protocol.
- **Path-vector:** Advertises AS_PATH; reject if own ASN appears.
- **Policy + attributes:** LOCAL_PREF, MED, AS_PATH, NEXT_HOP, etc.

### BGP message flow (5 stages)

1. **TCP setup:** Session on TCP/179.
2. **OPEN:** Exchange ASN, hold-time, capabilities.
3. **Initial UPDATE:** Send NLRI + attributes.
4. **Decision:** Import policy + best-path selection.
5. **Steady state:** KEEPALIVE + incremental UPDATE/NOTIFICATION.

### iBGP, eBGP and IGP interaction

- **eBGP:** Between ASes; increments AS_PATH, may change NEXT_HOP.
- **iBGP:** Within AS; distributes reachability to *external* prefixes and maps them to an egress point.
- **IGP:** Computes shortest paths *within* the AS and provides reachability to BGP NEXT_HOP / egress routers.
- **Best-path inside AS (hot-potato):** (1) Highest LOCAL_PREF, then (2) shortest AS_PATH, then (3) closest egress (lowest IGP cost), then (4) tie-break.

### BGP Stability

- **Stable state:** Routes converge with no further changes.
- **Dispute Wheel:** A cycle of nodes $u_0, \ldots, u_{k-1}$ where each $u_i$ prefers a path through $u_{i+1}$ over a direct/stable path.
- **Safety theorem (informal):**
  - **General:** BGP is safe (converges) iff the system contains no Dispute Wheel.
  - Under Gao-Rexford, BGP converges to a stable outcome (BGP-safe, no Dispute Wheel exists).

#### Gao-Rexford conditions (commercial routing)

  - **Topology:** No customer-provider cycles in the AS graph.
  - **Preference:** Prefer routes from customers over peers/providers.
  - **Export:** Advertise customer routes to all; peer/provider routes only to customers.

### BGP Security

- **Trust-based, weak by default:** BGP updates are accepted unless filtered; vulnerable to mistakes and attacks.
- **Session security (TCP-based):** eavesdropping, spoofing, reset, or tampering with UPDATEs.
- **Prefix hijacking:** False origin claims or more-specific prefixes attract traffic (can be accidental or malicious).
- **Detection:** Monitor multiple vantage points, anomaly alerts, route collectors/looking glasses.
- **Origin authentication:** Using a secure database to validate which AS may announce a prefix (reduces hijacks but not full path validation).

## Cross-Domain Subjects (Exam Oriented)

### Protocol influence on path selection:

| Protocol | Metric / Mechanism | Minimizes Weight? | Reasoning |
|---|---|---|---|
| STP | Loop Prevention (Tree) | No | Blocks physical links to prevent loops. Traffic must follow the tree, even if a direct (shorter) physical link exists. |
| OSPF | Cost (Bandwidth) | Yes | Uses Dijkstra's algorithm to mathematically guarantee the path with the lowest total cost. |
| RIP | Hop Count | Yes | Uses Bellman-Ford to find the path with the fewest hops (minimizes $weight = 1$ per link). |
| BGP | Policy (Attributes) | No | Prioritizes business rules (e.g., Local Preference) over path length. A longer "Customer" path is preferred over a shorter "Provider" path. |
| ECMP | Load Balancing | N/A | Mechanisms that split traffic across multiple paths *after* the routing protocol has already identified them as having equal (minimal) weight. |

### General tips and tools:

- mili=$10^{-3}$, micro=$10^{-6}$, nano=$10^{-9}$, kilo=$10^3$, mega=$10^6$, giga=$10^9$.
- In STP protcol problems, don't forget to document the Router, and who knows its location.
- When filling an outgoing message table (e.g. protocol messages), remember DHCP is in broadcast, ARP is needed to resolve MAC addresses (and is in Layer 2), and notice what LAN is followed.
- **You can't refer to MAC/IP addresses you don't know yet!**
- In Reliable Data Transfer Protocols, $T_{out}$ is also considered a parameter!
- When sending a packet with additional validity data (2D parity, checksum, CRC), this data isn't part of the effective payload (but is part of the total bits sent).
- In Traffic Engineering, try to locate the bottleneck using cuts; consider using the commodities vertices;
- In BGP we can define multiple IP prefixes for one AS, and give different policies (or priorities) to each prefix, from a different AS, hence splitting traffic based on prefix as well.